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战术资产配置的宏观经济仪表盘

——精品文献解读系列（三）

本报告导读：

本篇报告解读的文献构建了一套宏观经济仪表盘，可以帮助投资者解读宏观经济数据、筛选有效宏观因子，以更好地进行战术资产配置。

摘要：

- 投资学是一门学术界与业界紧密结合的学科，其中大类资产配置是这种紧密结合的代表。从 Markowitz（1952）开创现代投资组合理论开始，学术界为业界提供了丰富的理论参考和方法模型，推动了大类资产配置实践的繁荣发展。为了帮助读者及时跟踪学术前沿，我们推出了“精品文献解读”系列报告，从大量学术文献中挑选出精品论文进行剖析解读，为读者呈现大类资产配置领域最新的思路和方法。
- 本篇为读者解读的文献是 Clewell et al.（2018）在期刊 The Journal of Portfolio Management 上发表的论文“Macroeconomic Dashboards for Tactical Asset Allocation”。
- 该文献构建了一套宏观经济仪表盘，可以帮助投资者洞悉宏观因子对资产收益的影响。与其他文献不同，本文并没有对宏观因子的市场定价进行实证统计检验，而是重点强调了如何使用宏观因子数据来辅助进行投资策略。宏观经济仪表盘可以结合历史数据和对宏观因子的预期走势，对配对交易资产在未来 1 年的收益率进行预测，为投资者做出战术资产配置决策提供有益参考。

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1. 文献概述

文献来源：

Clewell, David, et al. "Macroeconomic Dashboards for Tactical Asset Allocation." Journal of Portfolio Management. 44.2 (2018): 50-61.

文献摘要：

本文构建了一套宏观经济仪表盘，可以帮助投资者洞悉宏观因子对资产收益的影响。与其他文献不同，本文并没有对宏观因子的市场定价进行实证统计检验，而是重点强调了如何使用宏观因子数据来辅助进行投资策略。宏观经济仪表盘可以结合历史数据和对宏观因子的预期走势，对配对交易资产在未来1年的收益率进行预测，为投资者做出战术资产配置决策提供有益参考。

文献评述：

如何根据宏观经济数据对资产配置进行中短期的战术调整，是所有投资者面临的重要问题，也是本文构建宏观经济仪表盘的目的。值得注意的是，本文并没有给出一套完整的基于宏观因子的战术配置策略，而仅仅是提供了一套解读宏观经济数据、筛选有效宏观因子的方法和思路。投资者可以参考本文构建自己的宏观经济仪表盘，并将从中得到的结论融入到自己的配置策略和投资决策中。

2. 引言

本文旨在建立一套宏观经济仪表盘（Macroeconomic Dashboards），以便于投资者将宏观因子（Macro Factors）应用于战术资产配置。无论是在学术界还是在业界，大家普遍认为宏观因子是资产收益率的重要驱动力。但是，很少有文献资料去阐释如何使用这些宏观因子来进行投资决策。在这种背景下，本文构建了一套宏观经济仪表盘，方便投资者从大量宏观数据中发现关键指标，进而更好地将宏观因子应用于战术资产配置。

本文不会去构建一套完整的基于宏观因子的大类资产配置策略，而是希望投资者能够结合宏观经济仪表盘与其他策略方法，更好地进行战术资产配置。将宏观因子融入传统配置策略至关重要。比如，传统的基于相对估值的战术资产配置策略，通常简单地根据估值指标（比如PE）进行“低买高卖”，但是这种策略在宏观经济剧烈变动时期可能会失效。本文构建的宏观经济仪表盘能够及时跟踪宏观因子，并对资产的未来业绩表现进行预判，可以帮助投资者将宏观因子融入战术资产配置策略。

2.1. 宏观因子对资产收益有重要影响

大量学术文献验证了宏观因子对资产收益的重要影响。Chen et al.(1986)发现，按照规模分类的股票投资组合对宏观因子（如利率、工业生产、通货膨胀、信用利差、消费等）的敏感性可以在很大程度上解释投资组

合之间的业绩差异。Fama and French (1989) 通过研究股票和债券市场，发现可以使用利率、信用利差等宏观因素来预测资产收益率。其他的大量研究也都验证了宏观因子在解释大类资产表现和风格溢价上的重要性。表 1 列举了近些年来相关文献所研究的宏观因子，主要包括消费、失业率、通过通胀、经济增长和石油价格等。

表 1：现有文献对宏观因子的研究（部分）

Jensen, Johnson, and Mercer [1997] show that the size and value effects depend largely on the monetary environment.
Booth and Booth [1997] emphasize the importance of monetary policy in explaining stock and bond returns.
Lettau and Ludvigson [2001] show how consumption patterns forecast excess equity market returns.
Dahlquist and Harvey [2001] show that the yield curve is predictive of stock returns.
Zhang et al. [2009] find a significant relationship between macroeconomic variables and the size and style premia.
They show that value and small-cap stocks outperform in periods of economic growth. Also, they find a positive relationship between inflation and value outperformance.
Boyd, Hu, and Jagannathan [2005] analyze the effect of unemployment and find that rising unemployment is bad news for stocks during contractions but, counterintuitively, good news during expansions.
Ludvigson and Ng [2009] show that “real” and “inflation” factors have predictive power for bond returns.
Kritzman, Turkington, and Page [2012] show that regimes in inflation and GDP explain asset class returns and that macro regime-switching models can be used for tactical asset allocation.
Franz [2013] shows that macro factors can be used to dynamically optimize an investor’s stocks vs. bonds allocation over time.
Page [2013] shows a significant link between macroeconomic surprises (in GDP and inflation) and bond, stock, and commodity returns.
Naik et al. [2016] relate the business cycle to returns for a broad list of risk factors.
Bernanke [2016] explains that oil prices and the stock market are correlated (albeit not perfectly) because of a common exposure to aggregate demand.
Blair and Qiao [2017] build a market-timing model for U.S. stocks with 20 factors, including oil, inflation rates, and spreads.

数据来源：Clewett et al. (2018)，国泰君安证券研究

2.2. 在资产配置中应用宏观因子存在困难

虽然学术界一再强调宏观因子的重要性，但是投资者在进行战术资产配置（6-18 个月）时却难以将宏观因子融入配置策略。这是因为学术界很少去讨论如何将宏观预期进行市场定价。此外，宏观经济数据种类繁多指标冗杂，这使得投资者难以区分其中的噪声和信号，无法找到驱动资产收益的有效指标。

在资产配置实践中面临的另外一个问题是，宏观因子在不同的初始条件下，对资产收益的影响可能大不相同。比如，Boyd et al. (2005) 的研究表明，失业率上升对股票收益的影响在经济扩张期和衰退期是完全不同的；类似地，工业生产下降对股票收益的影响也取决于初始经济环境的好坏。我们认为宏观因子对资产收益的影响取决于当前所处的情境状态，但这一问题在先前的学术研究中鲜有讨论。

3. 构建宏观经济仪表盘

本文通过构建宏观经济仪表盘，将宏观因子映射到资产预期收益。与其它学术论文通常采用计量经济学方法不同，本文使用的方法更加简便，更加直观。我们构建的宏观经济仪表盘具备以下特点：

（1）并不是基于静态历史数据样本进行回归分析，而是会进行动态更新，可以为投资者进行战术资产配置提供实时的参考；

- (2) 关注配对资产之间的相对收益情况;
- (3) 强调在不同情境状态下, 有哪些宏观因子对资产配对交易有显著影响;
- (4) 通过宏观因子当前所处的百分位水平来反应当前所处经济状态。

本文构建的宏观经济仪表盘包含了学术文献关注的主要宏观因子, 具体如表 2 所示。

表 2: 宏观经济仪表盘包含的宏观因子

Macro Factor	Change Type (Level, %)	Factor Start Date
VIX	% Level	Jan. 1990
Global Industrial Production ex-Construction (Y/Y)	Level	Jan. 1991
Oil Price (Brent Crude)	%	Jan. 1990
U.S. Dollar Index	%	Jan. 1990
High-Yield Credit Spread (Spread-to-Worst)	% Level	Jan. 1990
U.S. Investment-Grade Credit Spread (Option-Adjusted Spread)	% Level	Jan. 1990
U.S. Manufacturing PMI	Level	Jan. 1990
Real 10-Year U.S. Treasury Yield	Level	Jan. 1990
Real 2-Year U.S. Treasury Yield	Level	Jan. 1990
U.S. 10-Year vs. U.S. 2-Year Nominal Yield Differential	Level	Jan. 1990
Gold	%	Jan. 1990
U.S. Core Inflation (Y/Y)	Level	Jan. 1990
Euro Core Inflation (Y/Y)	Level	Jan. 1990
Euro Spot	%	Jan. 1990
Yen Spot	%	Jan. 1990
U.S. Unemployment Rate	Level	Jan. 1990
J.P. Morgan Emerging Markets Currency Index	%	Jan. 2000

Notes: All series are retrieved from Haver, except for high-yield spreads (J.P. Morgan Global High Yield Spread-to-Worst), J.P. Morgan Emerging Markets Currency Index, DXY (Factset), and U.S. investment-grade spreads (Bloomberg Barclays U.S. Aggregate Index OAS). Historical analysis data end in December 2016. All data are sourced at the monthly frequency. Levels are based on data reported April 10, 2017. We estimate real yields as the nominal Treasury yield less year-over-year Core CPI.

数据来源: Clewell et al. (2018), 国泰君安证券研究

表 3 展示了我们关注的配对交易资产 (asset-class-level pair trades)。对于每一对资产, 我们基于当前宏观因子的水平和情境状态, 通过回顾历史数据, 试图回答如下问题: 如果投资者能够预测宏观因子未来一年的变动方向, 那么持有该配对资产在未来一年可以获得的收益率是多少?

表 3：资产类别回报

Stocks vs. Bonds	Russell 3000 vs. Bloomberg Barclays U.S. Aggregate	Jan-1990
Stocks vs. Global High Yield	Russell 3000 vs. Credit Suisse High Yield	Jan-1990
Int'l Equity vs. U.S. Equity	MSCI ACWI ex-U.S. vs. Russell 3000	Jan-1990
EM Equity vs. DM Equity	MSCI EM vs. MSCI EAFE	Jan-1990
U.S. Growth vs. U.S. Value	Russell 1000 Growth vs. Russell 1000 Value	Jan-1990
DM Growth vs. DM Value	MSCI EAFE Growth vs. MSCI EAFE Value	Jan-1990
U.S. Small-Cap vs. U.S. Large-Cap	Russell 2000 vs. Russell 1000	Jan-1990
Int'l Small-Cap vs. Int'l Large-Cap	S&P Global Small Cap ex-U.S. vs. MSCI ACWI ex-U.S.	Jan-1997
Real Assets vs. Global Equities	T. Rowe Price Real Assets Blended Benchmark vs. MSCI ACWI	Jan-1990
Global High Yield vs. U.S. IG	Credit Suisse High Yield vs. Bloomberg Barclays U.S. Aggregate	Jan-1990
EM Dollar-Debt vs. U.S. IG	J.P. Morgan EMBI Global vs. Bloomberg Barclays U.S. Aggregate	Jan-1993
Non-\$ Debt vs. U.S. IG	Bloomberg Barclays Global Aggregate ex-U.S. vs. Bloomberg Barclays U.S. Aggregate	Jan-1990
EM Dollar-Debt vs. Global High Yield	J.P. Morgan EMBI Global vs. Credit Suisse High Yield	Jan-1993
Long Euro/Short USD	EURUSD (Spot)	Jan-1990
Long USD/Short Yen	USDJPY (Spot)	Jan-1990
Long Euro/Short Yen	EURJPY (Spot)	Jan-1990
Long USD/Short DM Currencies	U.S. Dollar Index (DXY; Spot)	Jan-1990
Long EM Currencies/Short USD	J.P. Morgan Emerging Markets Currency Index (Spot)	Jan-2000

Notes: Historical analysis data end in December 2016. Unless specifically identified, asset class returns are computed using total return indices. All data are sourced monthly. The T. Rowe Price Real Assets Blended Benchmark is the following: As of December 1, 2013, the Real Assets Combined Index Portfolio comprises 25% MSCI ACWI Metals & Mining, 20% Wilshire RESI, 20% FTSE EPRA/NAREIT Dev Real Estate Index, 19.5% MSCI ACWI Energy, 10.5% MSCI ACWI Materials, 4% MSCI ACWI IMI Gold, and 1.00% MSCI ACWI IMI Precious Metals and Minerals. Prior to this date, the Real Assets Combined Index Portfolio was composed of 25% MSCI ACWI Metals & Mining, 20% Wilshire RESI, 20% FTSE EPRA/NAREIT Dev Real Estate Index, 16.25% MSCI ACWI Energy, 8.75% MSCI ACWI Materials, 5% UBS World Infrastructure and Utilities Index, 4% MSCI ACWI IMI Gold, and 1.00% MSCI ACWI IMI Precious Metals and Minerals.

Sources: Bloomberg Barclays, Russell, Credit Suisse, FactSet, J.P. Morgan, and T. Rowe Price.

数据来源：Clewett et al. (2018)，国泰君安证券研究

图 1 展示了构建宏观经济仪表盘的思路方法。为了构建宏观经济仪表盘，首先定义宏观因子 f 的初始状态 (initial conditions) $IC(f)_t$ ，其取值可能为 *low*、*medium* 或 *high*，如式 (1) 所示：

$$IC(f)_t = \begin{cases} \text{"low"} & x_L \leq f_t < x_{(25\%*f)} \\ \text{"medium"} & x_{(25\%*f)} \leq f_t \leq x_{(75\%*f)} \\ \text{"high"} & x_{(75\%*f)} < f_t \leq x_U \end{cases} \quad (1)$$

其中， f_t 表示宏观因子 f 在时间 t 的取值； x_U 和 x_L 分别表示宏观因子 f 在历史上的最大值、最小值； $x_{(25\%*f)}$ 和 $x_{(75\%*f)}$ 分别表示宏观因子 f 在历史时间序列中 25% 分位数、75% 分位数。

接下来，我们定义宏观因子 f 在时间 $t+1$ 的情境指标 (scenarios) $S(f)_{t+1}$ ，如式 (2) 所示：

$$S(f)_{t+1} = f_{t+1} - f_t \quad (2)$$

结合情景指标 $S(f)_{t+1}$ 的数值，投资者需要根据经验定义宏观因子 f 的预期表现，包括“平稳” (Stable)、“上升” (Rising) 和“下降” (Declining)。比如，如果预期宏观因子 10 年期国债收益率从 t 到 $t+1$ 时刻的涨幅高于 25bp，则可定义该宏观因子的预期表现为“上升”。

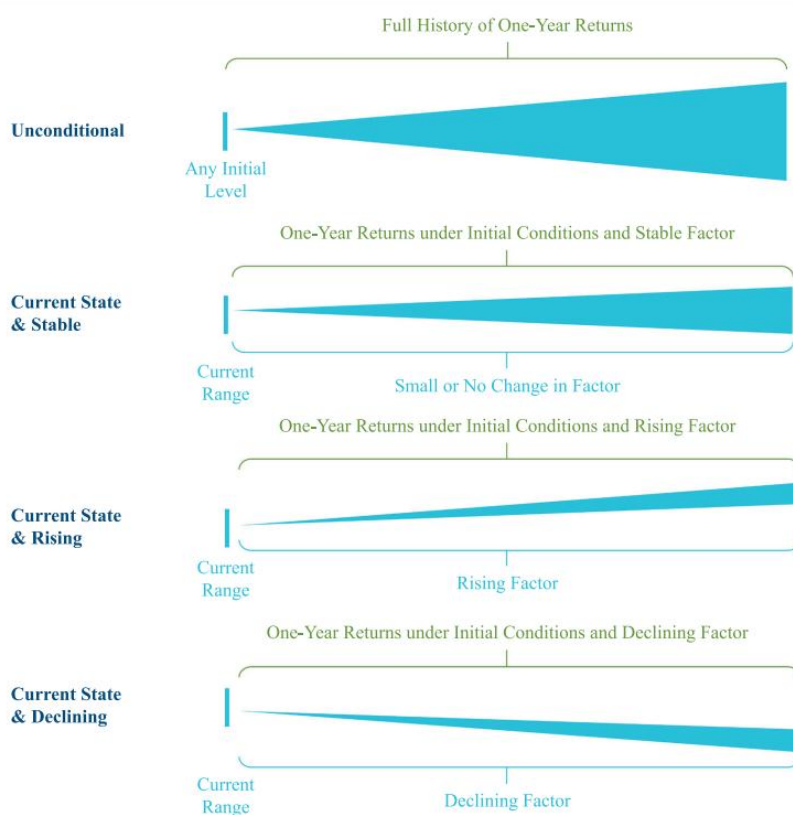
最后，计算配对交易的条件期望收益率 R_{t+1}^C ，如式（3）所示：

$$R_{t+1}^C = \mathbb{E}(R_{t+1} | IC_t, S_{t+1}) \quad (3)$$

具体来说，我们寻找在历史上出现过的同时匹配 IC_t 和 S_{t+1} 的时期，计算这些情境下相应配对交易收益率的平均值，即为 R_{t+1}^C 。同时，我们在宏观经济仪表盘中还列示了相应情境下收益率 10%分位数和 90%分位数。

比如，我们想要评估宏观因子美元，对配对交易资产——小盘股与大盘股——的影响。从整个样本区间来看，在 1990 年 11 月至 2016 年 12 月期间，美国小盘股（罗素 2000 指数）12 个月滚动收益率跑赢大盘股（罗素 1000）的概率为 51%。从当前时间节点（2017 年 4 月 10 日）来看，最新的美元指数达到 100.6，处于 1990 年 11 月以来的前 25%分位。进一步地，假设投资者在进行战术资产配置时，预期美元指数在未来会进一步上涨。回顾历史数据，当美元处于前 25%分位，并且在随后 1 年继续上涨 5%（或更多）时，美国小盘股收益率在随后 1 年跑赢大盘股的概率是 88%，平均超额收益是 8.2%，10%和 90%分位超额收益分别是 -2.3% 和 15.9%。综上，当美元处于高位且预期会继续上涨时，小盘股收益大概率能够跑赢大盘股，其背后的原因可能是美元在高位继续上涨影响了出口贸易，而小盘股受出口的影响比大盘股较小。

图 1：构建宏观经济仪表盘的思路



数据来源：Clewett et al. (2018)，国泰君安证券研究

4. 解读宏观经济仪表盘

表 4 展示了宏观经济仪表盘的示意图，包括对区域 1-区域 6 的解释。其中最重要的数据栏是区域 6 中的“条件收益率”（Conditional Returns），该条件收益率由所有宏观因子条件收益率加权得到。

表 4：宏观经济仪表盘示意图

Macro Factor Analysis: 12-Month Forward Return Outlook Given Current State
(as of April 10, 2017)

Macro Factor	Performance Over Entire Time Period		Conditional Returns		Global IP ex-Construction	³ U.S. Mfg. PMI	U.S. Unemployment ⁴
Current Value					0.79%	54.8%	4.5%
Long-Term Percentile Rank State	Avg Return % & [Range] (A)	Hit Rate (B)	Avg Return % & [Range] (C)	Hit Rate (D)	34 (Medium)	70 (Medium)	11 (Low)
Russell 3000 vs. U.S. Agg ¹	4.7 [-23, 22.5]	72% ²	3.6 [-23.9, 22.3]	69% ⁶	7.1 [-9.4, 22.4]	6.1 [-4.6, 17]	⁵ -4.1 [-35, 16.8]
Russell 3000 vs. Credit Suisse High Yield	1.3 [-18.1, 17]	58%	1.7 [-16.9, 16.9]	61%	5 [-7.7, 17.1]	5.3 [-3, 14.8]	0.6 [-19.2, 18.4]
MSCI ACWI ex-U.S. vs. Russell 3000	-4.1 [-19.9, 12.3]	36%	-2.7 [-16.6, 12.4]	39%	-2.9 [-16.4, 12.4]	-5.3 [-20.5, 11.1]	-0.8 [-11.9, 12.1]
MSCI EM vs. MSCI EAFE	4.8 [-17.3, 27.9]	58%	3.3 [-16.2, 25.6]	55%	3.6 [-14.4, 22.9]	-3.9 [-18.3, 18.4]	3.4 [-32.6, 26.8]
Russell 1000 Growth vs. Russell 1000 Value	-0.5 [-12.9, 11.4]	51%	-1.1 [-13.7, 11.9]	48%	1.6 [-10.4, 13.8]	-0.3 [-12.6, 8.8]	-0.9 [-28.3, 22.9]

¹ Asset class decision with corresponding indices.

² For reference: Average 12M relative return from January 1990 through December 2016, 10th to 90th range, and positive return hit rate.

³ Five macro and five financial factors where investors may have stronger views.

⁴ Each factor includes the current value, the long-term percentile rank, and the current state. Low: 0–25th %tile; Medium: 25th to 75th %tile; High: 75th to 100th %tile.

⁵ Given each indicator and current state, calculate the average relative returns, the 10th to 90th %tile range, and color whether this variable (at least present state) has an 80% chance of indicating profits (green) or losses (red).

⁶ Aggregate average of returns and ranges. The p-value indicates whether the average aggregate return is statistically different than the unconditional (full sample) average; formally, it should be below 0.05 (or 95% confidence level) to indicate a meaningful difference.

数据来源：Clewett et al. (2018)，国泰君安证券研究

表 5 至表 8 分别展示了宏观因子在无预期、预期稳定、预期上升和预期下降的情况下，宏观经济仪表盘给出的结果。对比这些表格可以发现，宏观因子的变动方向会影响相关资产在未来 1 年的收益表现。

当宏观经济保持稳定或上涨时，风险资产的表现往往会比较优异，比如小盘股、高收益率债券、新兴市场债券等。另一方面，当美国失业率从低位开始上升时，往往会伴随着股票的下跌抛售。

新兴市场货币也是一个值得重点关注的因子。新兴市场货币保持稳定或上升时期，新兴市场股票、房地产、债券等均能获得较好的收益。当前新兴市场货币处在历史低位（底部 5%分位），如果后期大幅上涨或继续下跌，势必会导致相关资产的剧烈波动。此外，油价如果从当前水平（63%分位）保持稳定或者上涨，也会显著驱动新兴市场股票、房地产和债券的价格上涨。

从表 5 至表 8 中，我们还可以解读出很多其他的有用结论。总的来看，这些结论与经济常识、学术研究保持一致。宏观经济仪表盘中的条件收益率及其置信区间、历史胜率（hit rate）等指标可以帮助投资者从海量的宏观数据中快速找出有参考价值的宏观因子。同时值得注意的是，宏观经济仪表盘需要定期更新，因为初始条件的改变会导致投资结论的变化。

表 5：预测未来 1 年收益率的宏观经济仪表盘（基于当前宏观状态，对宏观因子变化不做预期）

Macro Factor	Performance Over Entire Time Period		Conditional Returns		Global IP ex-Construction	U.S. Mfg. PMI	U.S. Unemployment	Oil Price	U.S. Core Inflation	High-Yield Spread (Level Chg)	VIX (% Change)	Real 2Y U.S. Treasury Yield (Core)	U.S. Dollar Index	JPM EM Currency Index
Current Value					0.79%	54.8%	4.5%	USD \$52.2	2.22%	457 bps	10.6	-1.0%	100.6	68.2
Long-Term % Rank State	Avg Return & [Range] (A)	Hit Rate (B)	Avg Return & [Range] (C)	Hit Rate (D)	34 (Medium)	70 (Medium)	11 (Low)	63 (Medium)	46 (Medium)	34 (Medium)	2 (Low)	21 (Low)	85 (High)	5 (Low)
Russell 3000 vs. U.S. Agg	4.7 [-23, 22.5]	72%	3.6 [-23.9, 22.3]	69%	7.1 [-9.4, 22.4]	6.1 [-4.6, 17]	-4.1 [-35, 16.8]	2.1 [-24.3, 21.8]	1.9 [-31.2, 25.6]	4.5 [-21.1, 22]	8.3 [-1.6, 20.9]	10.3 [-3.7, 26.5]	-2.7 [-30.8, 21.5]	3.4 [-25.5, 28.4]
Russell 3000 vs. Credit Suisse High Yield	1.3 [-18.1, 17]	58%	1.7 [-16.9, 16.9]	61%	5 [-7.7, 17.1]	5.3 [-3, 14.8]	0.6 [-19.2, 18.4]	-3 [-20.9, 9.6]	1.4 [-21.9, 19.4]	3.2 [-10.5, 16.8]	5.7 [-3.5, 14.9]	3.2 [-7.7, 14]	-1.8 [-23.4, 18.8]	-3.1 [-22.9, 11.1]
MSCI ACWI ex-U.S. vs. Russell 3000	-4.1 [-19.9, 12.3]	36%	-2.7 [-16.6, 12.4]	39%	-2.9 [-16.4, 12.4]	-5.3 [-20.5, 11.1]	-0.8 [-11.9, 12.1]	-0.3 [-19.6, 14.6]	-3 [-18, 12.1]	-4.2 [-16.4, 11.5]	0 [-19.3, 17]	-7.4 [-15.9, 0.1]	-2 [-11.6, 10.4]	-0.7 [-11.7, 12.2]
MSCI EM vs. MSCI EAFE	4.8 [-17.3, 27.9]	58%	3.3 [-16.2, 25.6]	55%	3.6 [-14.4, 22.9]	-3.9 [-18.3, 10.4]	3.4 [-32.6, 26.8]	10 [-12.2, 29.2]	1.3 [-25.8, 23.8]	1.7 [-15.2, 26.6]	5.6 [-15.6, 31.2]	0 [-18.8, 16.7]	2.3 [-29.1, 23]	7 [-13, 21.7]
Russell 1000 Growth vs. Russell 1000 Value	-0.5 [-12.9, 11.4]	51%	-1.1 [-13.7, 11.9]	48%	1.6 [-10.4, 13.8]	-0.3 [-12.6, 8.8]	-0.9 [-28.3, 22.9]	-5.1 [-17.9, 9.5]	-0.8 [-14.6, 16.4]	-1.5 [-14.1, 9.4]	0.6 [-9.6, 8.1]	1.4 [-7.5, 8.1]	-3.2 [-31.8, 23.2]	-1.1 [-10, 7.2]
MSCI EAFE Growth vs. MSCI EAFE Value	-2.1 [-11.3, 7]	38%	-2.1 [-11.8, 7.8]	38%	-1 [-10.1, 8.2]	-1.6 [-9.7, 5.9]	-2.2 [-16.6, 9]	-3.4 [-12.4, 8]	-2.3 [-10.2, 5.8]	-2.5 [-12.6, 7.5]	-2.1 [-12.5, 7.7]	1.2 [-6, 8.7]	-4.9 [-19.4, 6.3]	-1.2 [-11.9, 8.9]
Russell 2000 vs. Russell 1000	0 [-9.7, 12.6]	51%	0.2 [-9.7, 12.8]	52%	-1.3 [-10.9, 11.5]	-2.3 [-9.4, 6.6]	-1.5 [-23.2, 12.9]	3 [-7.9, 16.1]	-0.6 [-15, 12.8]	1.2 [-8.3, 10.2]	-1.7 [-8.8, 6.5]	-0.5 [-8.6, 8.7]	1.8 [-24, 17.4]	3 [-8.1, 18.2]
S&P Global Small Cap ex-U.S. vs. MSCI ACWI ex-U.S.	3.1 [-5.1, 9.2]	71%	2.9 [-5, 8.7]	71%	1.9 [-5.7, 8.3]	3 [-2.3, 7.5]	0.1 [-6.6, 6.8]	5.3 [-1.3, 14]	1.9 [-6, 8.4]	2 [-4.5, 7.2]	1.8 [-4.6, 5.8]	4.3 [-1.6, 9.8]	3.6 [-5.7, 11.8]	7.6 [1.9, 15.8]
Real Assets Blended Benchmark vs. MSCI ACWI	1 [-15.7, 18]	52%	2 [-15.9, 20.4]	53%	-0.2 [-14.5, 15.9]	0.4 [-14.6, 15.4]	3.9 [-18.7, 23.6]	8.9 [-6.8, 22.7]	0.6 [-20.3, 22.5]	-0.9 [-15.4, 15.6]	-0.4 [-12.3, 14.7]	-3.4 [-18.9, 15.9]	5.5 [-18.7, 25]	5.9 [-14.2, 18.4]
Credit Suisse High Yield vs. Barclays U.S. Agg	3 [-9.3, 12.9]	66%	1.5 [-10.7, 10.7]	57%	1.6 [-7, 10.1]	0.4 [-7.3, 8.2]	-5.1 [-14.2, 5.5]	4.7 [-10.8, 21.8]	0.1 [-12.6, 10.2]	0.9 [-11.5, 9.3]	2.2 [-4.9, 7.3]	6.8 [-5.3, 18.4]	-1.4 [-12.6, 17.3]	6.1 [-6.3, 22.7]
JPM EMBI Global vs. Barclays U.S. Agg	4 [-10.8, 22.3]	64%	4.2 [-9.6, 21.5]	66%	7.2 [-8.5, 24.7]	1.2 [-9.4, 10.3]	-0.5 [-17.3, 17.6]	6.1 [-5, 21]	3.1 [-12.1, 21.4]	5.8 [-6.7, 26.5]	2.1 [-14.4, 27.3]	4.7 [-3.5, 14.3]	2.7 [-14, 22.3]	7.7 [-1.8, 21.3]
Barclays Global Agg ex-U.S. vs. Barclays U.S. Agg	-0.4 [-10.7, 9.4]	50%	-1.3 [-13.6, 9.1]	47%	-1.8 [-12.8, 8.1]	-1 [-14.5, 8.7]	-3 [-15.4, 6.4]	0.1 [-12.8, 11.2]	-1.8 [-13.3, 9.4]	-2.3 [-14, 7]	0.7 [-10.9, 8.7]	-2.7 [-12.6, 4.9]	-0.2 [-15.8, 13.2]	4 [-11.3, 15.3]
JPM EMBI Global vs. Credit Suisse High Yield	1.5 [-13.3, 18.8]	53%	2.6 [-11.2, 19.6]	58%	5.2 [-4, 21.7]	0.4 [-10.2, 11.2]	4.2 [-8.6, 21.1]	2.2 [-7.5, 12.7]	2.6 [-11.3, 16.8]	5 [-11.6, 22.9]	-0.3 [-19, 21.9]	-2.5 [-13.8, 5.7]	3.7 [-11.3, 21.1]	1.2 [-8.2, 9]
EURUSD (Spot Index)	-0.1 [-13.3, 13.8]	48%	-0.3 [-14.2, 14]	49%	-2.5 [-15, 12.7]	-0.4 [-13, 12.3]	0 [-13.6, 12.5]	1.8 [-12.6, 15.9]	0.3 [-13.6, 15.6]	-3.5 [-14.6, 7.2]	0.8 [-14.8, 12.3]	-3.4 [-15.5, 5.8]	2.9 [-11.7, 18]	5.1 [-14.8, 21.7]
USDJPY (Spot Index)	-0.5 [-13.1, 15.3]	43%	0.2 [-13.1, 16.2]	48%	0.8 [-13.6, 16.1]	0.6 [-12, 15.4]	-1.5 [-14.2, 14.1]	-1.6 [-12.3, 12]	1.3 [-13.5, 19.7]	0.8 [-13.6, 20.8]	2.5 [-11.8, 16.5]	3.9 [-10.8, 22]	-2.5 [-13.1, 13.7]	-3.6 [-12.2, 11]
EURJPY (Spot Index)	-0.9 [-18.5, 14.1]	49%	-0.4 [-18.5, 14.2]	52%	-2.1 [-19.6, 10.3]	-0.4 [-12.1, 9.7]	-1.4 [-22.3, 14.1]	-0.1 [-16.2, 11.9]	1.4 [-19.8, 16.6]	-2.7 [-21.5, 15.8]	2.7 [-9.5, 13.4]	0.2 [-16.2, 25.3]	0.1 [-19.2, 14.9]	0.5 [-12.4, 11.4]
U.S. Dollar Index (DXY)	0.7 [-10.9, 11.7]	54%	1 [-10.9, 12.7]	55%	2.6 [-9.4, 14.1]	0.8 [-9.4, 12.8]	0.7 [-9.5, 11.8]	-0.9 [-12.1, 10.8]	0.8 [-11.2, 11.9]	3.5 [-6.1, 13.4]	0.3 [-9.5, 14.1]	3.7 [-4.2, 16]	-1.6 [-13.1, 9.1]	-3 [-15.4, 14.1]
JPM EMCI	-2.5 [-14.3, 8.8]	42%	-3.1 [-14.8, 8.2]	38%	-3.3 [-16.5, 7.7]	-2.6 [-16.5, 6.8]	-3.4 [-14.4, 8.3]	0 [-12.3, 10.3]	-2.9 [-13.9, 8.3]	-5.4 [-15.9, 5.6]	-2.8 [-16.5, 9.3]	-5 [-16, 5.8]	-3.6 [-13, 8.3]	-2.3 [-16.3, 10.8]

At least 80% Positive Hit Rate At least 80% Negative Hit Rate

The P-value associated with (C) ≠ (A) is generally between 0.9 and 1.0 given current economic factors. This means the current average and distribution is not statistically different from the long-term averages.

注：表 5-表 8 中的当前时刻指的均是 2017-04-10。

数据来源：Clewett et al. (2018)，国泰君安证券研究

表 6：预测未来 1 年收益率的宏观经济仪表盘（基于当前宏观状态，假设宏观因子未来 1 年预期保持稳定）

Macro Factor	Performance Over Entire Time Period	Global IP ex-Construction	U.S. Mfg. PMI	U.S. Unemployment	Oil Price	U.S. Core Inflation	High-Yield Spread (Level Chg)	VIX (% Change)	Real 2Y U.S. Treasury Yield (Core)	U.S. Dollar Index	JPM EM Currency Index
Current Value		0.79%	54.8%	4.5%	USDS 52.2	2.22%	457 bps	10.6	-1.0%	100.6	68.2
Long-Term Percentile Rank State	Avg Return & [Range]	34 (Medium)	70 (Medium)	11 (Low)	63 (Medium)	46 (Medium)	34 (Medium)	2 (Low)	21 (Low)	85 (High)	5 (Low)
Within the Following Range		100 bps	100 bps	25 bps	10%	25 bps	50 bps	5%	25 bps	5%	10%
Russell 3000 vs. U.S. Agg	4.7 [-23, 22.5]	9.1 [-2.7, 20.5]	10.9 [3.8, 15.1]	5.8 [-13.4, 15.9]	1.6 [-25.1, 20]	2.5 [-24.5, 22.6]	11.8 [0.5, 22.4]	9.6 [1.9, 21]	7.5 [-3.1, 20.7]	1.8 [-26.9, 21.6]	-3.9 [-32, 19.2]
Russell 3000 vs. Credit Suisse High Yield	1.3 [-18.1, 17]	3.6 [-5.1, 13]	6.1 [1.2, 11.1]	8.6 [-1.5, 19.2]	-3 [-15.8, 7.7]	4 [-17, 19.2]	8.3 [-2.5, 18]	5.5 [-2.2, 14.2]	1.9 [-20.1, 12.5]	4.7 [-20.6, 23.2]	-8.8 [-26.2, 4.4]
MSCI ACWI ex-U.S. vs. Russell 3000	-4.1 [-19.9, 12.3]	1 [-13.5, 14.4]	6.3 [-4.9, 16.5]	-1.3 [-11.9, 13.2]	-3 [-20, 11]	-5.4 [-18.4, 10.6]	-7 [-19.7, 9]	1.2 [-17.3, 16.8]	-6.7 [-15.5, 2]	-8.2 [-17, -0.5]	0.4 [-9, 11.3]
MSCI EM vs. MSCI EAFE	4.8 [-17.3, 27.9]	6.7 [-8.6, 23.3]	9.7 [-4.3, 21.8]	0.7 [-26.4, 26.1]	9.6 [-4.5, 24.4]	-1.9 [-37.8, 23.7]	10 [-3.5, 37.5]	6.8 [-22.6, 31.3]	1.1 [-14.5, 15.6]	-5.1 [-39.6, 23.2]	8.3 [-2.5, 17.7]
Russell 1000 Growth vs. Russell 1000 Value	-0.5 [-12.9, 11.4]	-2.5 [-12.1, 5.9]	-0.4 [-6, 5.5]	1.4 [-19.2, 21.8]	-7.8 [-22.9, 1.3]	-1.8 [-25.1, 16.9]	3.9 [-3.9, 18.6]	-0.6 [-8.9, 6.4]	3.1 [-4.1, 9.4]	1 [-22.4, 19.5]	-3.9 [-11.9, 1.8]
MSCI EAFE Growth vs. MSCI EAFE Value	-2.1 [-11.3, 7]	-2.7 [-10.7, 7.3]	-2.4 [-7.8, 3.9]	-3.6 [-19.1, 6.5]	-5.6 [-17.8, 1.9]	-1.9 [-10.8, 8.6]	-1.6 [-11.2, 6.7]	-5 [-12.9, 2.3]	0.9 [-6.3, 8.2]	-2.7 [-11.1, 7.1]	-1.7 [-11.4, 7.9]
Russell 2000 vs. Russell 1000	0 [-9.7, 12.6]	0.6 [-8.8, 16.3]	1.5 [-3.7, 6.8]	-7.3 [-29.2, 5.6]	2.5 [-10.1, 16.9]	-2.8 [-27.9, 14.3]	-2.5 [-9.5, 3.2]	-0.6 [-6.5, 6.7]	-2.6 [-9.5, 4.2]	-7.9 [-30, 13.2]	2.6 [-7.6, 16.3]
S&P Global Small Cap ex-U.S. vs. MSCI ACWI ex-U.S.	3.1 [-5.1, 9.2]	4.5 [-1.4, 9]	4 [2.1, 5.9]	-0.6 [-6.9, 4.6]	5.9 [2.2, 13.4]	0.7 [-7.1, 6.2]	2.2 [-3.2, 6.3]	3.5 [0.2, 6.2]	2.2 [-2.4, 8.2]	0.8 [-7, 7.6]	6.9 [3.1, 11.9]
Real Assets Blended Benchmark vs. MSCI ACWI	1 [-15.7, 18]	4.8 [-9.3, 15.9]	4.7 [-8.2, 14.7]	0.2 [-18.7, 15.6]	10.1 [-0.7, 23]	-0.2 [-23.5, 24.2]	-1.6 [-13.2, 16.4]	-1.4 [-11.8, 15.2]	-3.4 [-15.2, 14]	-1.2 [-23.3, 22.8]	11.4 [1.5, 18.4]
Credit Suisse High Yield vs. Barclays U.S. Agg	3 [-9.3, 12.9]	5.1 [-4.6, 17.4]	4.4 [0.3, 9]	-3.2 [-13.3, 6]	4.2 [-11.6, 19.8]	-1.9 [-12, 8.7]	3.1 [1, 4.8]	3.6 [-1.4, 7.3]	5.2 [-5.9, 14.4]	-3.2 [-12.1, 6]	4.5 [-6.1, 17.9]
JPM EMBI Global vs. Barclays U.S. Agg	4 [-10.8, 22.3]	8.5 [0, 22.4]	3.3 [-9.1, 10.1]	1.6 [-15.3, 14.4]	7.7 [-7.3, 22.4]	-1.3 [-19.1, 14.1]	13.7 [-7.6, 30.6]	-3.7 [-19.4, 6.9]	4.6 [-4.5, 13.4]	-4.6 [-22, 12.7]	7.9 [1.8, 21]
Barclays Global Agg ex-U.S. vs. Barclays U.S. Agg	-0.4 [-10.7, 9.4]	1.5 [-8, 10.3]	0.8 [-9.2, 8.8]	-3.2 [-16.4, 4.5]	1.2 [-13.8, 12.3]	-3.8 [-13, 6.5]	0.8 [-6.5, 9.1]	1.6 [-8.1, 8.3]	-3.4 [-13.5, 4.6]	-0.3 [-8.3, 5.2]	8 [1.5, 16.6]
JPM EMBI Global vs. Credit Suisse High Yield	1.5 [-13.3, 18.8]	3 [-3.3, 8.3]	-1.4 [-16.8, 8.2]	4.3 [-8.4, 21.5]	0.2 [-4.6, 15.8]	10.3 [-13, 12.9]	0.2 [-13.8, 27.2]	-7.8 [-23.4, 4.7]	-1 [-8.7, 5.7]	-3 [-14.8, 11.7]	3 [-6.6, 9.3]
EURUSD (Spot Index)	-0.1 [-13.3, 13.8]	-0.2 [-8.9, 13.4]	-0.9 [-10.1, 12]	-0.9 [-12.4, 7.6]	-1.9 [-9.9, 17]	-0.2 [-12.4, 9.5]	-0.1 [-10.2, 11.7]	1.9 [-8.6, 9.1]	-4.7 [-16.7, 5.2]	0.2 [-4.8, 5.5]	10.3 [-0.4, 24.5]
USDJPY (Spot Index)	-0.5 [-13.1, 15.3]	1.4 [-11.8, 13.9]	-0.6 [-13.1, 12]	-2.4 [-13.4, 8]	-1.5 [-12.9, 11.4]	4.5 [-12.8, 21]	0.9 [-17.1, 21.4]	1.5 [-10.6, 11.7]	3.3 [-9.3, 16.5]	-3.7 [-18.6, 12.7]	-7.2 [-15, 0.3]
EURJPY (Spot Index)	-0.9 [-18.5, 14.1]	3.3 [-13.4, 13]	-1.5 [-8.3, 4.5]	-3.2 [-19.4, 12.1]	2.6 [-15.1, 12.7]	2.2 [-15.2, 17.3]	0.6 [-20.2, 15.7]	3.1 [-7, 12]	-2 [-12.4, 7]	-3.2 [-20.5, 17.4]	2.4 [-15.1, 15]
U.S. Dollar Index (DXY)	0.7 [-10.9, 11.7]	-1.1 [-10.5, 8.7]	0 [-11.2, 9.4]	0.9 [-5.4, 10.1]	-2.8 [-12.7, 8.6]	2.7 [-5.9, 10.9]	0.3 [-10.1, 7.3]	-0.8 [-6.1, 9.2]	4.9 [-4.1, 18.9]	0.1 [-1.7, 2.5]	-7.3 [-16.4, 0.9]
JPM EMCI	-2.5 [-14.3, 8.8]	0.2 [-12.2, 9.7]	3.7 [-1.1, 8.1]	0.3 [-7.3, 8.5]	-0.4 [-8.8, 10]	-3.9 [-14.1, 7.4]	-1.5 [-9, 6.4]	-4.4 [-11.6, 2.1]	-6.3 [-18.1, 4.5]	-6.8 [-14.1, 0.3]	-0.7 [-7.4, 5.3]
At least 80% Positive Hit Rate At least 80% Negative Hit Rate											

数据来源：Clewett et al. (2018)，国泰君安证券研究

表 7：预测未来 1 年收益率的宏观经济仪表盘（基于当前宏观状态，假设宏观因子未来 1 年预期上升）

Macro Factor	Performance Over Entire Time Period	Global IP ex-Construction	U.S. Mfg. PMI	U.S. Unemployment	Oil Price	U.S. Core Inflation	High-Yield Spread (Level Chg)	VIX (% Change)	Real 2Y U.S. Treasury Yield (Core)	U.S. Dollar Index	JPM EM Currency Index
Current Value		0.79%	54.8%	4.5%	USDS 52.2	2.22%	457 bps	10.6	-1.0%	100.6	68.2
Long-Term Percentile Rank State	Avg Return & [Range]	34 (Medium)	70 (Medium)	11 (Low)	63 (Medium)	46 (Medium)	34 (Medium)	2 (Low)	21 (Low)	85 (High)	5 (Low)
Increasing by More Than the Following		100 bps	100 bps	25 bps	10%	25 bps	50 bps	5%	25 bps	5%	10%
Russell 3000 vs. U.S. Agg	4.7 [-23, 22.5]	14.3 [3.5, 25.1]	11.9 [5.8, 18.9]	-24.2 [-40.9, -12]	4.3 [-19.8, 20.6]	2 [-17.4, 11.4]	-8.3 [-37.1, 9]	7.4 [-3.6, 21.4]	20.2 [6.4, 32.8]	-7.4 [-25.5, 16.3]	32.2 [19.8, 45.1]
Russell 3000 vs. Credit Suisse High Yield	1.3 [-18.1, 17]	11.1 [1.2, 21.7]	6.4 [0.7, 10.6]	-13.2 [-22.9, -2.6]	-2.9 [-22.9, 8.5]	3.2 [-10.4, 16.5]	-0.7 [-15.1, 11.2]	6.5 [-1.3, 16.6]	6.5 [-5.2, 17.6]	0.3 [-16.9, 18.4]	2.7 [-9.5, 15.3]
MSCI ACWI ex-U.S. vs. Russell 3000	-4.1 [-19.9, 12.3]	-3 [-26, 15]	-8.3 [-15.3, -4.2]	-0.8 [-8.8, 9.2]	7.5 [-5.3, 18.3]	1 [-12.1, 16.7]	-9 [-14.9, -0.9]	-3.6 [-19.4, 14.4]	-8.8 [-13.7, 4]	-4.7 [-10.1, 6.4]	10.8 [5.4, 16.5]
MSCI EM vs. MSCI EAFE	4.8 [-17.3, 27.9]	5.8 [-13.7, 25.4]	1.5 [-7.1, 10.8]	9.1 [-9.5, 25.3]	15.1 [-2.5, 32.4]	-0.4 [-11.8, 10.1]	-4.1 [-13.8, 4.9]	4 [-14.7, 27.5]	-1.7 [-22.2, 2.4]	2.1 [-13.6, 23.8]	25.2 [17.9, 37.6]
Russell 1000 Growth vs. Russell 1000 Value	-0.5 [-12.9, 11.4]	5.7 [-9, 28.2]	2.8 [1.4, 4.3]	-10 [-41, 8.9]	-2.2 [-12.6, 11.8]	-1.5 [-15.2, 27.7]	0 [-37.8, 16.8]	3.4 [-3.7, 9.3]	-1.8 [-9.1, 5.5]	-1.8 [-44, 29.2]	-1.8 [-6.4, 2.8]
MSCI EAFE Growth vs. MSCI EAFE Value	-2.1 [-11.3, 7]	-2.1 [-10.6, 5.8]	1.6 [-2.4, 5.6]	-2.4 [-21.1, 10]	-2.9 [-11.3, 8.5]	-3.5 [-17.2, 7.3]	-0.3 [-21.5, 9]	0.7 [-6.1, 9]	-7.9 [-7.1, 6.3]	-7.9 [-26.2, 14.4]	-11.4 [-14.2, -8.9]
Russell 2000 vs. Russell 1000	0 [-9.7, 12.6]	-2.8 [-12.6, 6.2]	-0.4 [-10.9, 8.8]	5.5 [-7, 15]	3.6 [-5.3, 11.5]	1.6 [-7.5, 8.2]	-0.2 [-8.6, 8.6]	-3.6 [-8.9, 1.6]	5 [-1, 10.7]	8.2 [-2.3, 15.9]	16 [6.7, 25.3]
S&P Global Small Cap ex-U.S. vs. MSCI ACWI ex-U.S.	3.1 [-5.1, 9.2]	-0.1 [-5.3, 5.1]	4.1 [-0.8, 7.6]	0.3 [-7.1, 6.9]	5.3 [-4.4, 15.4]	0 [-6.2, 4.2]	0.1 [-5.2, 6.8]	0.3 [-6.3, 5.7]	6.1 [1.3, 14.7]	0.6 [-4.5, 5.2]	15.4 [13.1, 17.3]
Real Assets Blended Benchmark vs. MSCI ACWI	1 [-15.7, 18]	-1.9 [-14.2, 15.4]	2.3 [-7.1, 10.9]	14.4 [-6.2, 28.2]	13 [4.9, 22.4]	6.3 [-12.8, 19.7]	-2 [-14.8, 24.5]	-1.4 [-14.3, 12.6]	-5.8 [-20.9, 15.7]	7.4 [-14, 28.6]	13.4 [6.5, 22.7]
Credit Suisse High Yield vs. Barclays U.S. Agg	3 [-9.3, 12.9]	2.8 [-4.3, 7.8]	5 [-1.1, 10.7]	-11.4 [-23.6, -3.1]	6.8 [-8.1, 23.5]	-1.6 [-12.9, 7]	-8 [-16.4, -1.4]	0.6 [-7, 6.8]	13.3 [2.4, 35.9]	-8.1 [-13.8, -2.2]	29.1 [19.4, 42.3]
JPM EMBI Global vs. Barclays U.S. Agg	4 [-10.8, 22.3]	12.2 [-8.6, 27.7]	0.9 [-9.7, 7.6]	-6.1 [-14, 0.7]	8.7 [-1.1, 21]	6.7 [-1.7, 17.9]	1 [-7.3, 15]	5.3 [-9.8, 31.3]	7.6 [-3.6, 16.9]	20.6 [-7.3, 22.4]	20.6 [16.1, 24.7]
Barclays Global Agg ex-U.S. vs. Barclays U.S. Agg	-0.4 [-10.7, 9.4]	-3.5 [-12.9, 7.9]	-1 [-7.4, 5.1]	-3.5 [-15.6, 8.2]	2.5 [-10.7, 11.4]	-3 [-16.3, 6.4]	-8.3 [-16.3, 1.6]	0.2 [-13.7, 8.9]	-9.4 [-9.4, 3.5]	-14 [-18.4, -8.5]	9.5 [3.9, 13.2]
JPM EMBI Global vs. Credit Suisse High Yield	1.5 [-13.3, 18.8]	-9 [-9.6, 25.6]	-4.6 [-13.7, 2.9]	4.9 [-2.9, 11.9]	1.5 [-11.3, 10.6]	7.9 [-4.1, 23.3]	8.6 [1.1, 21.1]	4.3 [-12, 24.8]	-8.6 [-19.4, 0]	15.3 [3.8, 25.8]	-8.9 [-21.4, 1]
EURUSD (Spot Index)	-0.1 [-13.3, 13.8]	-5.2 [-14.4, 9.8]	-1.5 [-8.1, 6.9]	2.1 [-12.8, 15.1]	4.6 [-12.7, 18]	0.6 [-13.8, 12.3]	-8.7 [-18.1, 1.4]	0.7 [-16.3, 13.5]	-1.3 [-13.3, 7.5]	-10.1 [-16, -4.3]	12.1 [5.9, 17.3]
USDJPY (Spot Index)	-0.5 [-13.1, 15.3]	-0.6 [-14.5, 13.7]	-1.1 [-10, 14.9]	-0.3 [-14.5, 14.6]	-2.5 [-11.2, 9]	-2.1 [-13.8, 6.5]	1.3 [-12.7, 16.9]	4.5 [-1.7, 21.9]	10.5 [-8.6, 25.2]	4.6 [-11.7, 15.5]	-8 [-11.8, -3.8]
EURJPY (Spot Index)	-0.9 [-18.5, 14.1]	-6 [-22.8, 7.8]	-3 [-12.9, 5.6]	-1.5 [-21.9, 14.2]	1.5 [-14.1, 10.3]	-1.6 [-18.5, 11.3]	-7.8 [-20.8, 4.7]	4.2 [-3.6, 16.9]	9.4 [-16.6, 32.1]	-5.6 [-21.9, 8.5]	3 [-2.5, 8.2]
U.S. Dollar Index (DXY)	0.7 [-10.9, 11.7]	3.6 [-7.3, 12.6]	0.7 [-6.2, 8.9]	0.3 [-11.4, 12.7]	-3.6 [-13, 9.4]	0 [-9.4, 12.6]	8.4 [-1.2, 18]	0.8 [-10.4, 16.5]	2.4 [-4.3, 10.9]	8.9 [5.3, 12.9]	-10.6 [-13.1, -6.9]
JPM EMCI	-2.5 [-14.3, 8.8]	-1.2 [-4.2, 2.1]	-0.8 [-8.7, 4.9]	-6.4 [-16.8, 9.1]	3.5 [-4.3, 10.8]	-1.8 [-7.3, 1.9]	-10.4 [-17.5, -3.3]	-3.4 [-17.8, 10.7]	-2 [-12.1, 12.5]	-7.8 [-13.4, -4.9]	12 [10.2, 14.3]
At least 80% Positive Hit Rate At least 80% Negative Hit Rate											

数据来源：Clewett et al. (2018)，国泰君安证券研究

表 8：预测未来 1 年收益率的宏观经济仪表盘（基于当前宏观状态，假设宏观因子未来 1 年预期下降）

Macro Factor	Performance Over Entire Time Period	Global IP ex-Construction	U.S. Mfg. PMI	U.S. Unemployment	Oil Price	U.S. Core Inflation	High-Yield Spread (Level Chg)	VIX (% Change)	Real 2Y U.S. Treasury Yield (Core)	U.S. Dollar Index	JPM EM Currency Index
Current Value		0.79%	54.8%	4.5%	USD\$ 52.2	2.22%	457 bps	10.6	-1.0%	100.6	68.2
Long-Term Percentile Rank State	Avg Return & [Range]	34 (Medium)	70 (Medium)	11 (Low)	63 (Medium)	46 (Medium)	34 (Medium)	2 (Low)	21 (Low)	85 (High)	5 (Low)
Decreasing by More Than the Following		100 bps	100 bps	25 bps	10%	25 bps	50 bps	5%	25 bps	5%	10%
Russell 3000 vs. U.S. Agg	4.7 [-23, 22.5]	-1.5 [-39, 13.7]	4.8 [-5, 15.7]	11.9 [4.3, 22.1]	-0.6 [-24.4, 21.8]	1.3 [-38.4, 28.8]	12.4 [0.8, 24.5]	8.8 [2.5, 17.9]	-0.9 [-8.3, 8.7]	-3.7 [-32, 27.8]	1.3 [-3.6, 8.3]
Russell 3000 vs. Credit Suisse High Yield	1.3 [-18.1, 17]	0.7 [-11.8, 11.6]	5 [-3.1, 16.4]	10.5 [-0.7, 21.9]	-2.9 [-21.6, 18.4]	-2.1 [-24.6, 20.2]	4.9 [-5.5, 17]	4 [-5.8, 13]	0.1 [-3.6, 2.3]	-9.5 [-26, 10.9]	6.2 [1.6, 11.1]
MSCI ACWI ex-U.S. vs. Russell 3000	-4.1 [-19.9, 12.3]	-6.4 [-14.1, 8.7]	-7 [-22.6, 10.8]	-0.5 [-17.3, 12.9]	-9.6 [-27.5, 7]	-2.4 [-22.6, 12.1]	0.2 [-15.1, 15.2]	6.9 [-12.7, 26.5]	-12.6 [-18.4, -8.3]	5.9 [-3.2, 14.3]	-10.8 [-13, -9]
MSCI EM vs. MSCI EAFE	4.8 [-17.3, 27.9]	-1.3 [-14.8, 16.3]	-6.6 [-24.1, 8.4]	-1.3 [-44.4, 40]	2.9 [-14.4, 35]	5.4 [-20.8, 26.3]	4 [-18.8, 29]	7.8 [-14.7, 31.9]	0.1 [-7.3, 9.2]	9.5 [-2.4, 21]	-8.1 [-14.8, 5.3]
Russell 1000 Growth vs. Russell 1000 Value	-0.5 [-12.9, 11.4]	1.4 [-9.4, 9.5]	-0.6 [-12.8, 9.5]	8.3 [-11, 29.6]	-6.9 [-25.1, 9.7]	0.7 [-9.3, 10.7]	-4 [-13.7, 5.5]	-4.4 [-13.8, 3.3]	2.5 [-2.3, 6]	-4.3 [-12.7, 3.7]	6 [1.6, 9.2]
MSCI EAFE Growth vs. MSCI EAFE Value	-2.1 [-11.3, 7]	1.6 [-5.4, 9.5]	-1.8 [-10.6, 5.9]	-0.8 [-8.5, 6.2]	-2 [-12.2, 8.9]	-2.2 [-9, 2.8]	-4.3 [-11.2, 3.9]	-5.5 [-15.6, 0]	5.8 [2.5, 9.5]	-5 [-12.9, 2.1]	6.9 [3.3, 9.1]
Russell 2000 vs. Russell 1000	0 [-9.7, 12.6]	-1.7 [-9.9, 9.7]	-3.1 [-9.3, 5.3]	-5.1 [-27.2, 5.6]	2.6 [-10.2, 16.7]	0.8 [-9.7, 11.7]	3.1 [-6.6, 10.5]	0.4 [-8.8, 12.2]	4.4 [-8.5, 0]	-5.4 [-2.7, 21.1]	-4.5 [-9.4, 0.7]
S&P Global Small Cap ex-U.S. vs. MSCI ACWI ex-U.S.	3.1 [-5.1, 9.2]	0.7 [-7.8, 8]	2.7 [-2.5, 7.4]	0.5 [-5.9, 9.4]	4.8 [-0.1, 8]	4.1 [-4.5, 13.8]	4.1 [0.7, 7.5]	3.4 [2, 4.5]	2 [-2.4, 6.6]	8.7 [5.2, 16.1]	4.1 [-2.4, 7.9]
Real Assets Blended Benchmark vs. MSCI ACWI	1 [-15.7, 18]	-3.2 [-14.6, 16]	-0.5 [-17.9, 15.7]	-5.7 [-22.7, 17.5]	1.7 [-15.9, 24.7]	-1.3 [-18.7, 17.5]	0.1 [-18.1, 14.8]	3.1 [-9.1, 18.4]	1.1 [-8, 15.1]	10.5 [0.4, 22.5]	-12.1 [-17.7, -5.5]
Credit Suisse High Yield vs. Barclays U.S. Agg	3 [-9.3, 12.9]	-2.5 [-18.9, 10.3]	-0.7 [-8.5, 6.4]	0.9 [-8, 6.7]	2 [-12.1, 24.4]	3 [-16.1, 21.8]	7.1 [4.1, 9.9]	4.4 [0.1, 8.9]	-1.4 [-5.2, 3.3]	5.4 [-7.6, 21.9]	-5.3 [-7, -3.1]
JPM EMBI Global vs. Barclays U.S. Agg	4 [-10.8, 22.3]	1.2 [-13.3, 14.2]	0.9 [-8.9, 10.7]	4.5 [-20.9, 25.1]	-1.6 [-9.4, 5.7]	6 [-10.5, 24.6]	8 [-4.6, 26.7]	1.5 [-11.2, 10.3]	4.2 [1, 8]	6.2 [-10.9, 22.3]	-1.4 [-4.8, 0.5]
Barclays Global Agg ex-U.S. vs. Barclays U.S. Agg	-0.4 [-10.7, 9.4]	-3.2 [-13.5, 7.8]	-1.3 [-15.2, 8.4]	-2.2 [-9.4, 4.6]	-4.6 [-13, 4.5]	0.8 [-10.3, 15]	1.4 [-4.6, 7.5]	1 [-7.4, 6.7]	-0.5 [-8, 6.3]	9.8 [3.6, 16.5]	-8.8 [-13.9, -1.6]
JPM EMBI Global vs. Credit Suisse High Yield	1.5 [-13.3, 18.8]	3.5 [-3.1, 9.9]	1.2 [-10, 18.9]	3.2 [-13, 23.6]	2.6 [-4.4, 10.6]	2.7 [-11.6, 13.9]	0.2 [-13.7, 21.5]	-2.9 [-19.7, 7.3]	5.2 [3.3, 8.5]	0.5 [-8.9, 9.3]	3.5 [0.2, 7]
EURUSD (Spot Index)	-0.1 [-13.3, 13.8]	-4.3 [-18, 14.4]	-0.3 [-14.7, 12.3]	-1.7 [-13.9, 10.3]	-5 [-14.9, 5.7]	2.7 [-14.3, 20]	-0.4 [-10.6, 7.7]	-3.8 [-10.6, 12.3]	14.9 [-13.9, 1.6]	-11.8 [7.6, 24.4]	-11.8 [-18.8, -3.1]
USDJPY (Spot Index)	-0.5 [-13.1, 15.3]	1.6 [-14.2, 20.2]	0.9 [-11.7, 15.1]	-2.2 [-13.8, 16]	-0.2 [-14.9, 13]	-0.4 [-14.5, 20.2]	0.4 [-13.6, 23.5]	-1.2 [-14.8, 11.1]	-7.8 [-15.7, -0.5]	-6.4 [-11.6, 0.3]	7.8 [-5.2, 17.4]
EURJPY (Spot Index)	-0.9 [-18.5, 14.1]	-3.3 [-17.3, 11.7]	0 [-12, 10.2]	-3.4 [-23.1, 16.6]	-5.1 [-22.2, 14.6]	2.1 [-22.3, 21.4]	0.3 [-21.4, 25.3]	-1.5 [-19.6, 10.9]	-11.6 [-16.2, -6.6]	7.6 [-1, 15.2]	-5.5 [-9.9, -1.1]
U.S. Dollar Index (DXY)	0.7 [-10.9, 11.7]	5.2 [-11, 19.3]	0.9 [-9.4, 13.3]	1.1 [-7.4, 9.1]	4.9 [-2.4, 13.2]	-0.8 [-14.7, 13.6]	0.5 [-7.1, 6.8]	0.2 [-9.3, 9.1]	2.7 [-1.7, 10.9]	-10.9 [-16.4, -6.1]	11.9 [3.2, 19.4]
JPM EMCI	-2.5 [-14.3, 8.8]	-7.5 [-18.2, 7.9]	-4 [-17.3, 6.5]	-0.9 [-3.2, 1.5]	-10.6 [-14.8, -4.4]	-2.7 [-16.6, 10.1]	0.2 [-7, 6.8]	0.3 [-3.6, 3.4]	-6.9 [-14.2, 1.1]	1.3 [-7.2, 10.5]	-15.7 [-19.2, -11.9]

At least 80% Positive Hit Rate At least 80% Negative Hit Rate

数据来源：Clewel et al. (2018)，国泰君安证券研究

5. 附加说明

关于宏观经济仪表盘的构建和使用，有如下几点值得注意：

(1) 宏观经济仪表盘展示的是相关性，并没有去深入探究宏观因子如何驱动资产价格变化，其背后的因果关系需要投资者自行探寻。

(2) 本文没有直接使用预期数据进行建模，原因是市场预期通常难以获得和度量。调查数据往往不及时、不全面，而远期数据又难以从中剔除风险溢价。同时，宏观因子本身也难以获得准确预期，如 Chen et al. (1986) 认为利差和利率时间序列受噪音影响往往无法进行预测。

(3) 尽管大量学术论文认为宏观因子是驱动资产价格变动的重要因素，但是从宏观经济仪表盘中可以看到，对未来 1 年收益率预期的置信区间较宽，统计显著性也不高。因此，在构建系统性的战术资产配置策略时，我们不建议只考虑宏观数据，而是需要把宏观因子和其他因子（如相对估值因子、基本面因子、技术因子等）结合起来进行投资决策。比如，我们可以综合考虑宏观因子和相对估值因子：当相对估值因子（PE 或其他估值指标）显示新兴市场股票比美国股票便宜，且宏观因子也显示新兴市场股票预期会上涨时，那么战术资产配置者就应该增加新兴市场股

票配置比例。

（4）宏观经济仪表盘旨在从宏观因子层面为投资者提供参考和灵感，在具体进行战术资产配置时，除了需要结合其他因子，还要进一步进行其他相关处理，比如调整因子暴露程度、对冲配对资产的市场风险等。

6. 结论

回顾历史可以帮助我们更好地预测未来。结合历史和当前数据，本文构建了宏观经济仪表盘，可以帮助投资者洞悉宏观因子对资产收益的影响。与其他文献不同，本文并没有对宏观因子的市场定价进行实证统计检验，而是重点强调了如何使用宏观因子数据来辅助进行投资决策。宏观经济仪表盘可以结合历史数据和对宏观因子的预期走势，对配对交易资产在未来 1 年的收益率进行预测，为投资者做出战术资产配置决策提供有益参考。

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